



HikCentral Professional Video SDK (C++) Demo

User Manual

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Chapter 1 Overview

HikCentral Professional Video SDK (C++) demo (hereinafter referred to as demo) is developed based on Visual Studio 2015 and Qt with version 5.9.6. It provides API calling examples and functions of live view, playback, two-way audio, and event subscription.

The demo has configured with 32-bit or 64-bit compilation environment, and the executable file name is HCVideoSDKDemo.exe.

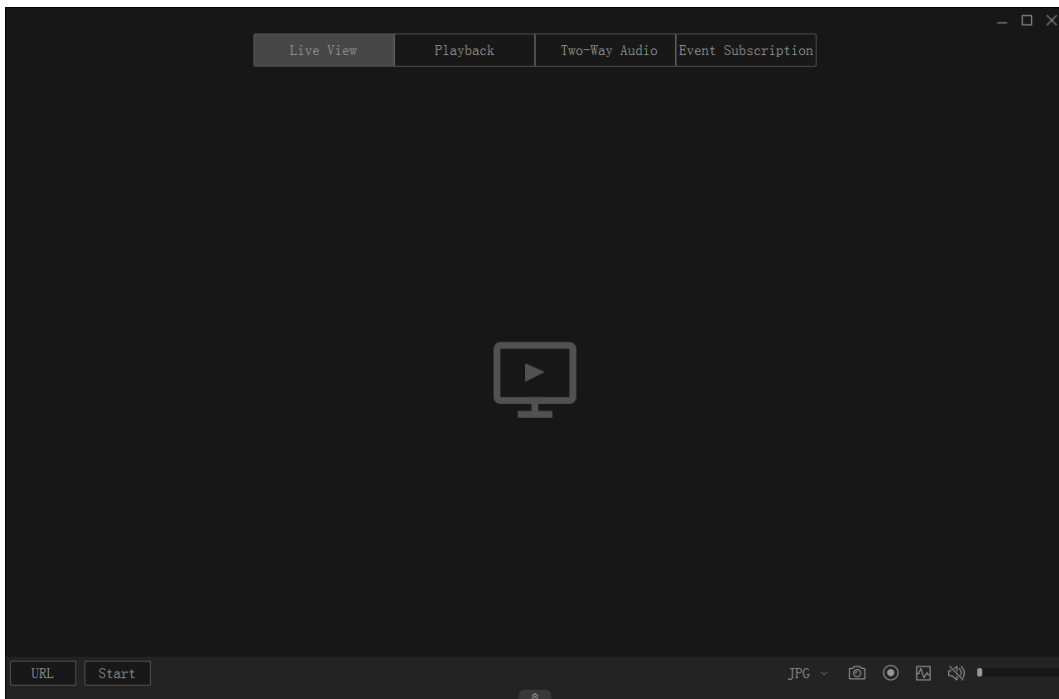
Chapter 2 Live View

Purpose:

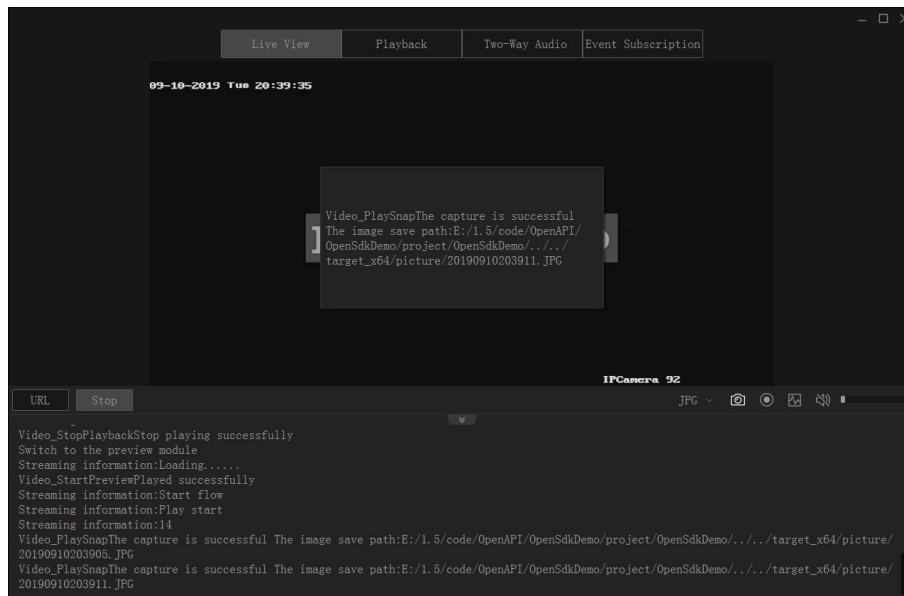
You can enter the URL information to start live view, and then stop live view. Before starting live view, you can choose to enable GPU hardware decoding and display video image or not. During the live view, capturing, recording, viewing video details, and audio control are supported.

Steps:

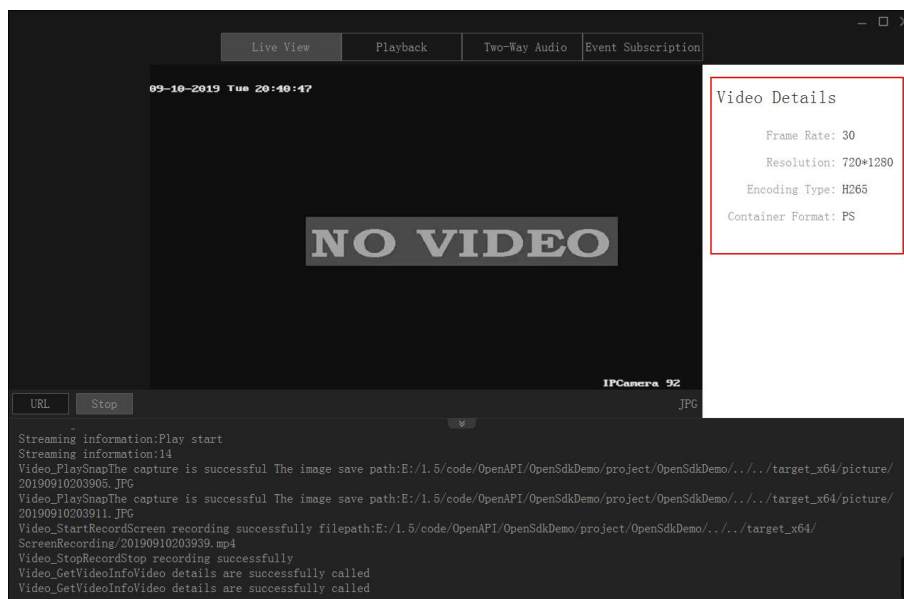
1. Run the *HCVideoSDKDemo.exe* in OpenAPI installation directory to open the 32-bit demo.
2. Click **Live View** tab to enter live view page.



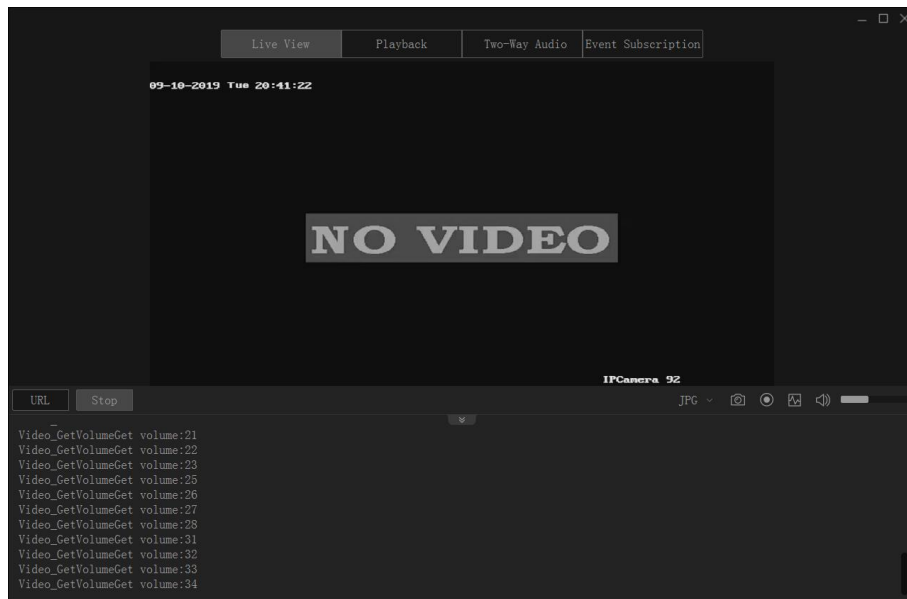
3. Click **URL** and configure the following parameters.
 - **URL:** Live view or streaming URL, which can be obtained by calling gateway API. Before starting each time of live view, you should get the live view URL based on RTSP.
 - **Authentication:** Authentication information.
 - **Open GPU:** Whether to enable GPU hardware decoding.
 - **Display:** Whether to display video image.
4. Click **Start** to start live view.
5. (Optional) Perform the following operation(s) after starting live view.
 - **Capture:** Click  to start capturing JPG or BMP2 picture. A window will pop up to show the capture results, including failure or success information and saving path of picture (an absolute path, and the same directory of *HCVideoSDKDemo.exe*).



- Recording: Click  to start recording MP4 video and click  again to stop. A window will pop up to show the recording results, including failure or success information and saving path of video files (an absolute path, and the same directory of *HCVideoSDKDemo.exe*). The recorded video must be played in Hikvision player.
- Video Details: Click  to view the video details.



- Audio Control: Click  to turn on audio and click  again to turn off audio. And you can drag on the volume bar to adjust the volume when the audio is on.



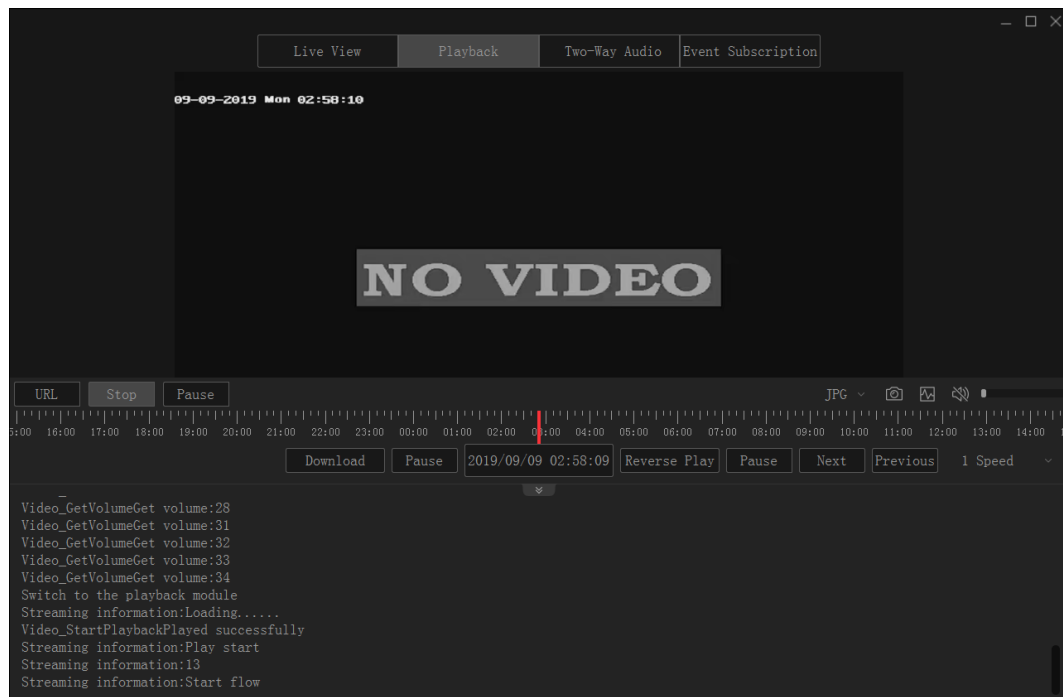
Chapter 3 Playback

Purpose:

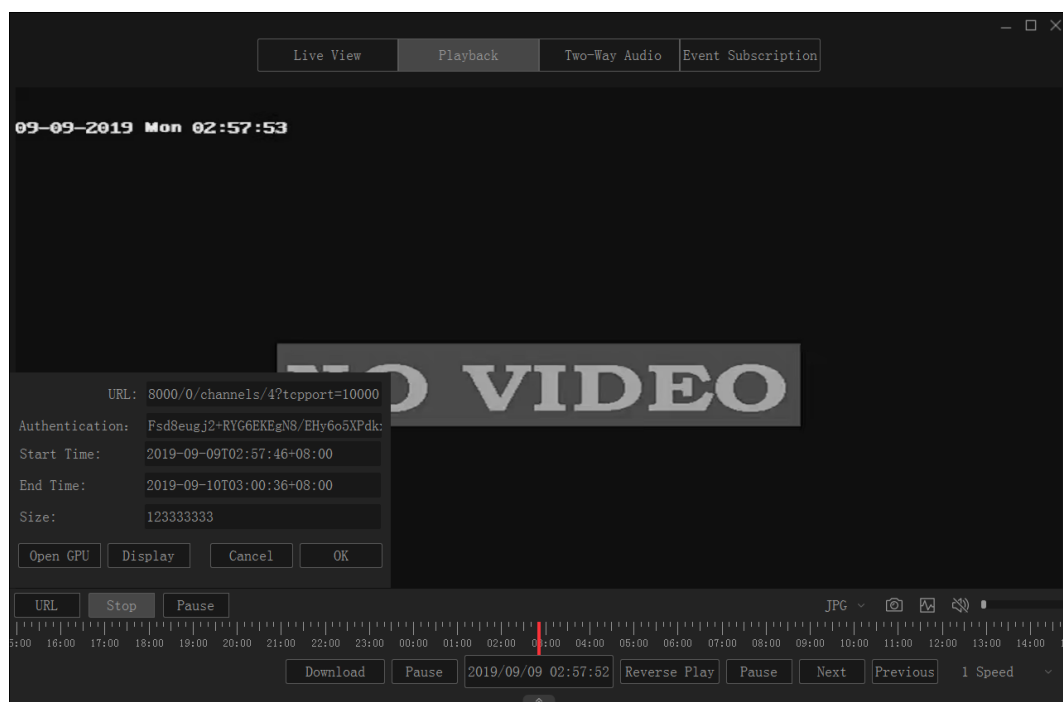
You can enter the URL information and set start/end time to start playback or reverse playback. Before starting playback, you can choose to enable GPU hardware decoding and display video image or not. During the playback, pausing or resuming playback, capturing, recording, viewing video details, audio control, fast or slow forward playback, single frame playback, and so on are supported.

Steps:

1. Run the *HCVideoSDKDemo.exe* in OpenAPI installation directory to open the 32-bit demo.
2. Click **Playback** tab to enter playback page.



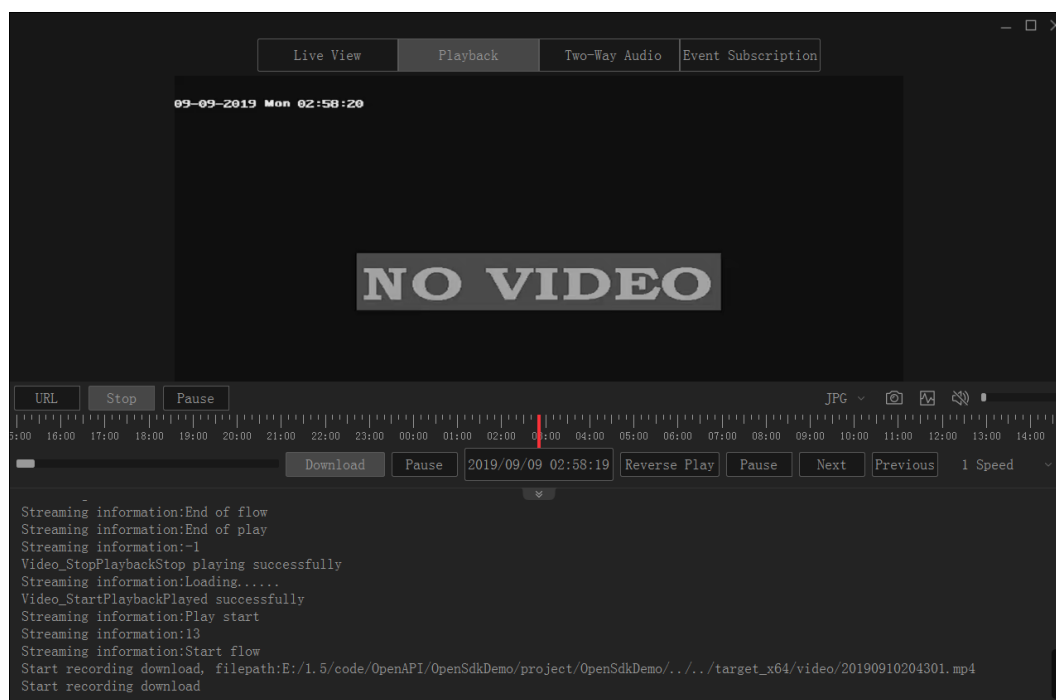
3. Click **URL** and configure the following parameters.



- **URL:** Playback or streaming URL, which can be obtained by calling gateway API. Before starting each time of playback, you should get the live view URL based on RTSP.
- **Authentication:** Authentication information.
- **Start Time:** Timestamp for starting playback (unit: second).
- **End Time:** Timestamp for stopping playback (unit: second).
- **Size:** Video segment size.
- **Open GPU:** Whether to enable GPU hardware decoding.
- **Display:** Whether to display video image.

Note:

- If the start time is earlier than the end time, it will start normal playback; if the start time is later than the end time, reserve playback will start; if the start time and end time are same, error will occur.
 - During normal playback, make sure the start time is later than the search start time of playback URL; during reverse playback, make sure the start time is earlier than the search end time of playback URL.
4. Click **Start** to start playback.
 5. (Optional) Perform the following operation(s) after starting playback.



- **Pause/Resume Playback:** Click **Pause** to pause the playback and click **Restore** to resume the playback from paused video position.
- **Capture:** Click  to start capturing JPG or BMP2 picture. A window will pop up to show the capture results, including failure or success information and saving path of picture (an absolute path, and the same directory of *HCVideoSDKDemo.exe*).
- **Recording:** Click  to start recording MP4 video and click  again to stop. A window will pop up to show the recording results, including failure or success information and saving path of video files (an absolute path, and the same directory of *HCVideoSDKDemo.exe*). The recorded video must be played in Hikvision player.
- **Video Details:** Click  to view the video details.
- **Audio Control:** Click  to turn on audio and click  again to turn off audio. And you can drag on the volume bar to adjust the volume when the audio is on.
- **Located Playback:** Click on the playback time bar to specify a video time point to start playing, or select date from calendar to play the videos of the day. You can edit when the playback stopped, and the selected time point must be between start time and end time.
- **Fast/Slow Forward Playback:** Select the playback speed to fast forward or slow forward.
- **Reverse Playback:** Click **Reverse Play** to reversely play the video in the specified time period. You can click **Pause** to stop reverse playback and click **Restore** to continue playing from the paused time point in normal speed.
- **Single Frame Playback:** Click **Previous** or **Next** to play the previous or next frame one by one.
- **Download Video:** Click **Download** to start downloading video according to URL. The downloading time duration is same as that of playback, and the video size is same as that in

the URL. You can click **Pause** to stop downloading and click **Restore** to continue downloading.

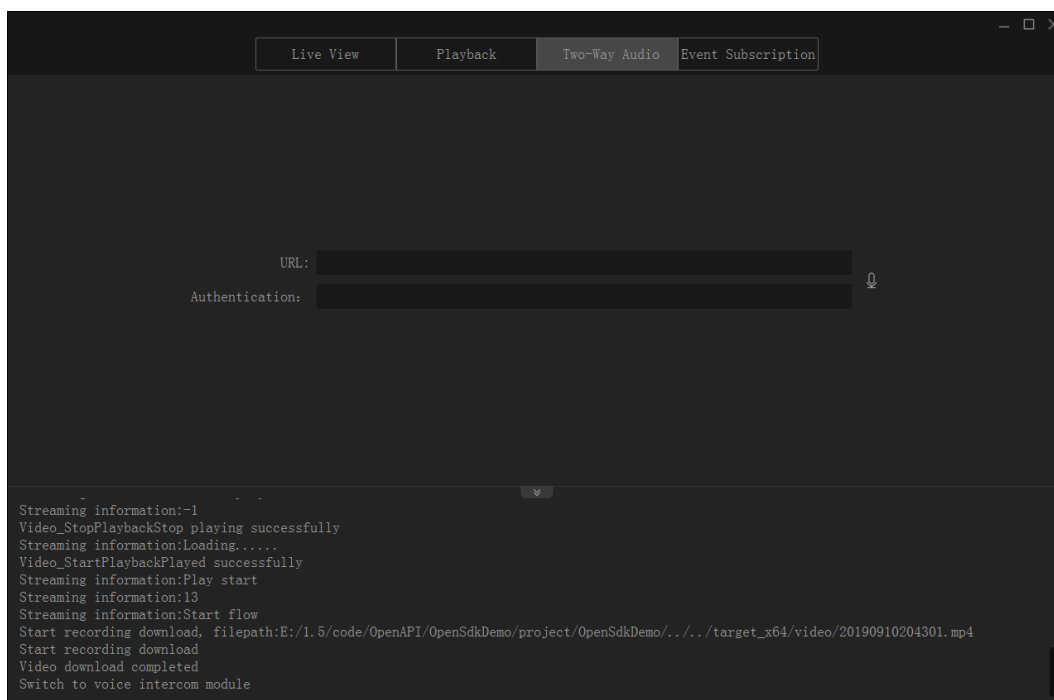
Chapter 4 Two-Way Audio

Purpose:

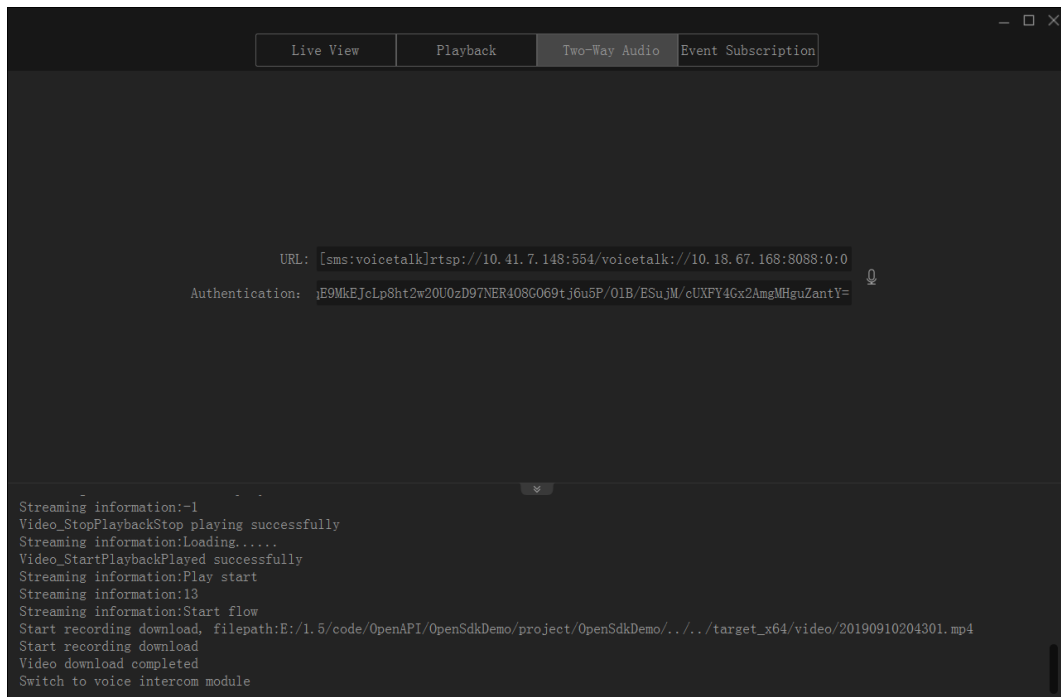
You can enter the URL information to start two-way audio with the device via the demo.

Steps:

1. Run the *HCVideoSDKDemo.exe* in OpenAPI installation directory to open the 32-bit demo.
2. Click **Two-Way Audio** tab to enter two-way audio page.



3. Click **URL** and configure the following parameters.
 - **URL:** Playback or streaming URL, which can be obtained by calling gateway API. Before starting each time of playback, you should get the live view URL based on RTSP.
 - **Authentication:** Authentication information.
4. Click  to start two-way audio.



The color of  will turn to red if started; otherwise, it indicates that starting two-way audio failed.

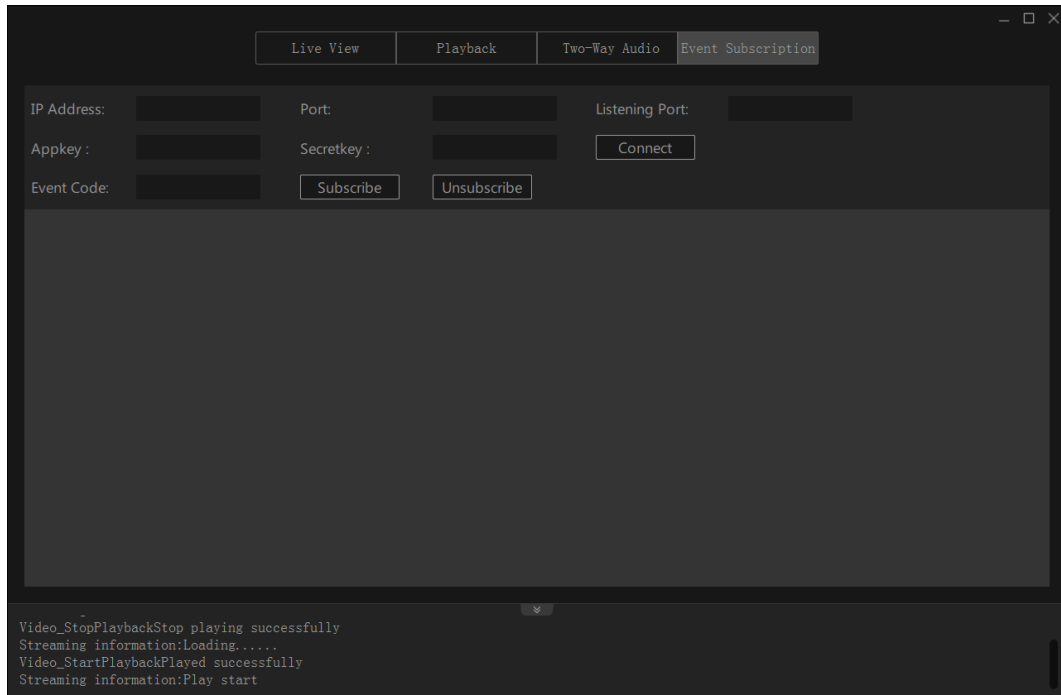
Chapter 5 Event Subscription

Purpose:

You can subscribe multiple event types according to the event codes to receive event information regularly via the demo.

Step:

1. Run the *HCVideoSDKDemo.exe* in OpenAPI installation directory to open the 32-bit demo.
2. Click **Event Subscription** tab to enter event subscription page.



3. Configure the following parameters.
 - **IP Address:** IP address of server, this address does not support HTTPS.
 - **Port:** Port No. of server.
 - **Listening Port:** Local listening port No.
 - **Appkey:** Third-party partner key.
 - **SecretKey:** Third-party partner secret key.
4. Click **Connect** to establish the connection.
5. Enter event code, such as 131331 (motion detection).
6. Click **Subscribe** to subscribe the event.
7. (Optional) Repeat *step 5* and *step 6* to subscribe multiple events.
8. (Optional) Click **Unsubscribe** to cancel subscribing the events.



See Far, Go Further